

$$\boxed{1.29} \quad \gamma_m = \int_0^1 t^m e^{-t} dt, \quad m=0, 1, 2, \dots$$

$$(a) \quad \gamma_0 = \int_0^1 e^{-t} dt = -e^{-t} \Big|_0^1 = -e^{-1} + e^0 = 1 - e^{-1}$$

$$\begin{aligned} \gamma_m &= \int_0^1 t^m e^{-t} dt = -\int_0^1 t^m (e^{-t})' dt = \\ &= -t^m e^{-t} \Big|_0^1 + m \int_0^1 t^{m-1} e^{-t} dt = -e^{-1} + m \gamma_{m-1} \end{aligned}$$

$$\Rightarrow \begin{cases} \gamma_k = k \cdot \gamma_{k-1} - e^{-1} \\ \gamma_0 = 1 - e^{-1} \end{cases} \quad k=1, 2, 3, \dots$$

$$\begin{aligned} (b) \quad \gamma_1 &= 1 \cdot (1 - e^{-1}) - e^{-1} = 1 - 2e^{-1} \\ \gamma_2 &= 2(1 - 2e^{-1}) - e^{-1} = 2 - 5e^{-1} \\ \gamma_3 &= 3(2 - 5e^{-1}) - e^{-1} = 3! - 16e^{-1} \\ \gamma_4 &= 4(3! - 16e^{-1}) - e^{-1} = 4! - 65e^{-1} \\ &\dots \Rightarrow \gamma_k = k! - x \cdot e^{-1} \end{aligned}$$

$k!$ este rapid crescătoare și de aceea vom dăa în $\gamma(20)$ un număr cu 18 cifre.

$$\begin{aligned} \gamma_{20} &= 20! - x e^{-1} \approx 8.95 e+17 \\ 20! &\approx 2.43 e+18 \end{aligned}$$

$$\begin{aligned} (c) \quad k \gamma_{k-1} &= \gamma_k + e^{-1} \\ \gamma_{k-1} &= \frac{1}{k} (\gamma_k + e^{-1}) \\ \gamma_k &= 0 \end{aligned} \quad \begin{aligned} \gamma_{k-1} &= \frac{e^{-1}}{k} \\ \gamma_{k-2} &= \frac{1}{k} \left(\frac{e^{-1}}{k} + e^{-1} \right) \end{aligned}$$